

Bond Pricing, Duration & Convexity

Price, duration and convexity — the three numbers every fixed-income desk lives by, verified against the worst bond year in modern history.

A bond is just a promise of future cash flows — so its price is entirely a question of *discounting*. From that single idea follow the three numbers every fixed-income desk lives by: **price**, **duration** (first-order rate sensitivity, the bond's Delta) and **convexity** (second-order, the bond's Gamma).

PROJECT CARD

CATEGORY

Quant Foundations & Derivatives

LIBRARIES

NumPy · Pandas · Matplotlib

ASSETS

SHY, IEF, TLT (US Treasury ETFs)

TIMEFRAME

2022 (rate-shock year)

USE

Measuring and stress-testing interest-rate risk in a bond book

1. Summary

A bond's price is the present value of its coupons and principal, discounted at the **yield to maturity**. Price and yield move inversely, and the relationship is *curved*, not linear. **Duration** measures the percentage price change

for a small yield change; **convexity** corrects that estimate for the curvature — and the correction always works in the bondholder's favour.

2. Intuition

2.1 Discounting, not magic

When market yields rise from 2% to 4%, an old bond paying 2% coupons is suddenly a bad deal — nobody pays face value for below-market cash flows. Its price falls until a buyer earns the *new* market yield. No sentiment involved: pure arithmetic.

2.2 Duration = weighted waiting time

Macaulay duration is the average time you wait for your money, weighting each cash flow by its share of present

value. A 30y bond makes you wait decades → huge sensitivity. A 2y note returns your money quickly → barely reacts. **Rule of thumb: a bond loses ~duration × Δy percent** when yields rise by Δy.

2.3 Convexity: the curvature that helps you

The price-yield curve is convex: losses from rising yields are *smaller* than the linear estimate, gains from falling yields are *larger*. Long bonds have the most convexity — which is why duration alone increasingly misleads for big yield moves.

3. Theory & Mechanics

With face F , coupon rate c (paid f times a year), yield y and $n = fT$ periods:

$$P = \sum_{i=1}^n \frac{cF/f}{(1 + y/f)^i} + \frac{F}{(1 + y/f)^n}$$

Macaulay duration (years): $D_{mac} = \frac{1}{P} \sum_i t_i \cdot PV(CF_i)$, and **modified duration** $D = D_{mac}/(1 + y/f)$.

Convexity C is the second derivative of price w.r.t. yield, scaled by price. Both derivatives can also be computed numerically — which is how we'll verify the analytic formulas:

$$\frac{\Delta P}{P} \approx -D \Delta y + \frac{1}{2} C (\Delta y)^2$$

```
def bond_price(face, coupon, y, T, freq=2):
    """Price of a fixed-coupon bond (coupon = annual rate, y = YTM)."""
    n = int(round(T * freq))
    t = np.arange(1, n + 1)
    cf = np.full(n, face * coupon / freq)
    cf[-1] += face
    return float(np.sum(cf / (1 + y / freq) ** t))

def duration_convexity(face, coupon, y, T, freq=2, h=1e-4):
    """Modified duration (years) and convexity via central differences."""
    p0 = bond_price(face, coupon, y, T, freq)
    p_up, p_dn = bond_price(face, coupon, y + h, T, freq), bond_price(face, coupon, y - h, T, freq)
    dur = -(p_up - p_dn) / (2 * h * p0)
    conv = (p_up - 2 * p0 + p_dn) / (h**2 * p0)
    return dur, conv

# quick check against the rule of thumb
```

```
p = bond_price(100, 0.04, 0.04, 10)
d, c = duration_convexity(100, 0.04, 0.04, 10)
print(f"10y 4% bond at par: P = {p:.2f}, duration = {d:.2f}y, convexity = {c:.1f}")
```

4. Applied Example — US Treasuries

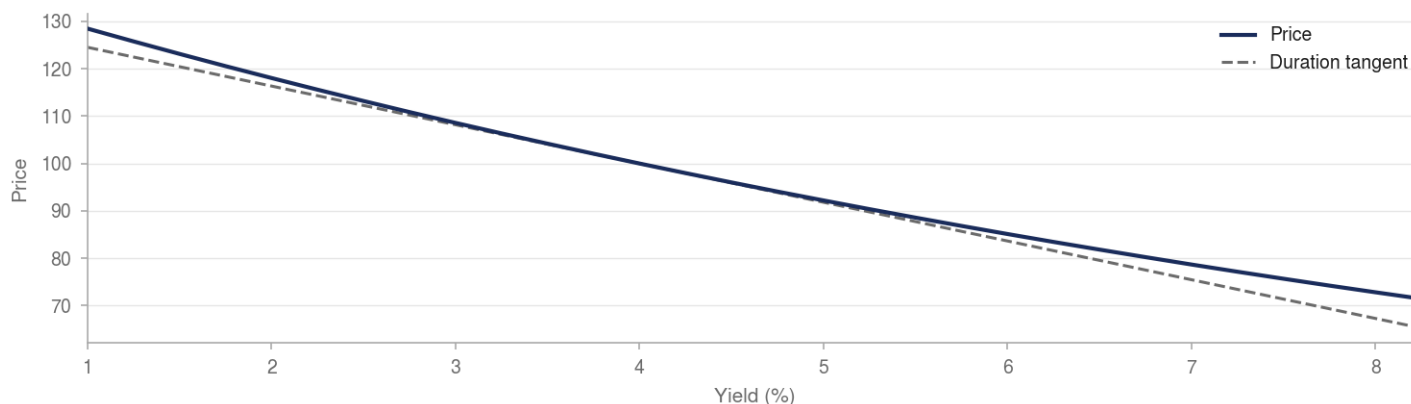
4.1 The price-yield curve, by maturity

Same 4% coupon, three maturities. Watch two things: the **slope** (longer = steeper = more duration) and the **curvature** (longer = more convex).

```
yields = np.linspace(0.001, 0.10, 200)
fig, ax = plt.subplots(figsize=(10, 5))
for T, color in [(2, "steelblue"), (10, "darkorange"), (30, "crimson")]:
    prices = [bond_price(100, 0.04, y, T) for y in yields]
    ax.plot(yields * 100, prices, color=color, label=f"{T}y Treasury")
ax.axhline(100, color="gray", ls=":", lw=1)
ax.axvline(4, color="gray", ls=":", lw=1)
ax.set_xlabel("Yield (%)"); ax.set_ylabel("Price")
ax.set_title("Price vs yield – 4% coupon, three maturities")
ax.legend()
plt.tight_layout(); plt.show()
```

Figure 4.1 · Price vs yield — 4% coupon, three maturities

The price curve against the straight-line duration estimate; the gap between them is convexity



Source: Author calculations, from the article's computed results.

4.2 The duration ladder: a +100bp shock

The whole risk story in one table — duration grows with maturity, and so does the damage from the same yield move.

```
rows = []
for T in [1, 2, 5, 10, 20, 30]:
    d, c = duration_convexity(100, 0.04, 0.04, T)
    p0 = bond_price(100, 0.04, 0.04, T)
    p1 = bond_price(100, 0.04, 0.05, T) # +100bp, exact
    rows.append([T, round(d, 2), round(c, 1), f"{(p1/p0 - 1)*100:.2f}%"])

ladder = pd.DataFrame(rows, columns=["Maturity (y)", "Mod. duration", "Convexity", "Exact dP for +100bp"])
print(ladder.to_string(index=False))
```

Maturity (y)	Mod. duration	Convexity	Exact dP for +100bp
1	0.97	1.4	-0.96%
2	1.90	4.6	-1.88%
5	4.49	23.5	-4.38%
10	8.18	78.9	-7.79%
20	13.68	239.9	-12.55%
30	17.38	420.8	-15.45%

4.3 Reality check: SHY, IEF, TLT in 2022

2022 was the worst bond year in modern history: the 2y yield rose ~370bp, the 10y ~237bp, the 30y ~207bp. If duration analytics mean anything, the three Treasury ETFs' losses should line up with their durations (~1.9 / ~7.5 / ~17.5). Let's check against real prices.

```
def load_prices(ticker, start, end):
    """Adjusted-close prices: yfinance first, Stooq as fallback."""
    try:
        import yfinance as yf
        df = yf.download(ticker, start=start, end=end, auto_adjust=True, progress=False)
        if not df.empty:
            return df["Close"].squeeze().rename(ticker).dropna()
    except Exception as exc:
        print(f"yfinance failed ({exc}); trying Stooq...")
        url = f"https://stooq.com/q/d/l/?s={ticker.lower()}.us&i=d"
        df = pd.read_csv(url, parse_dates=["Date"], index_col="Date")
        return df.loc[start:end, "Close"].rename(ticker).dropna()

etfs = {"SHY": (1.9, 0.0370), "IEF": (7.5, 0.0237), "TLT": (17.5, 0.0207)} # (duration, 2022 dY)

px = pd.concat([load_prices(t, "2021-12-31", "2022-12-31") for t in etfs], axis=1).dropna()
total_ret = px.iloc[-1] / px.iloc[0] - 1

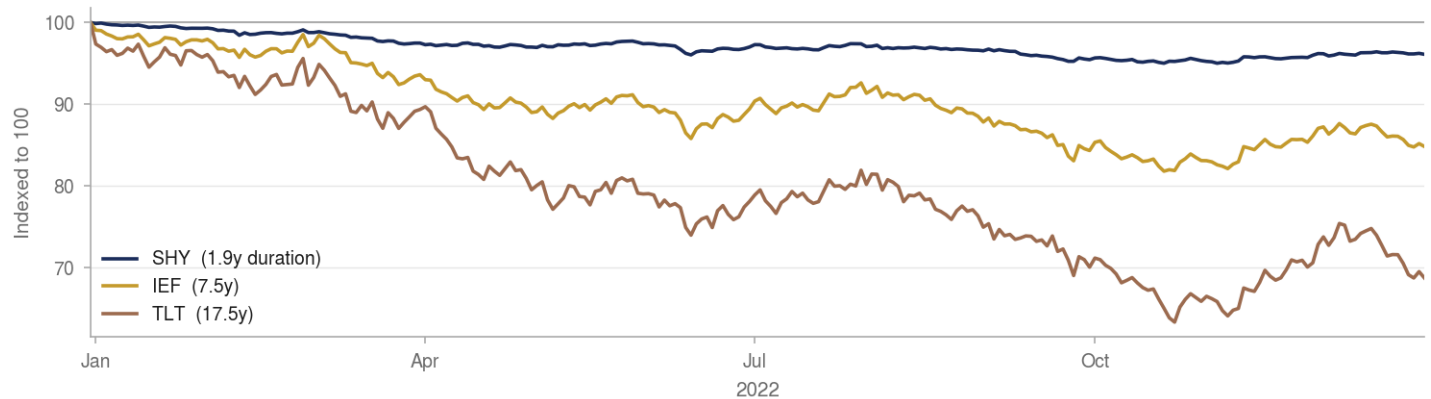
print(f"{'ETF':<4} {'duration':>9} {'dY 2022':>8} {'predicted -D*dY':>16} {'actual 2022':>12}")
for t, (d, dy) in etfs.items():
    print(f"{t:<4} {d:9.1f} {dy:8.2%} {-d*dy:16.1%} {total_ret[t]:12.1%}")

(px / px.iloc[0] * 100).plot(figsize=(10, 4), color=["steelblue", "darkorange", "crimson"])
plt.title("2022: the duration ladder in real life (indexed to 100)")
plt.ylabel("Value"); plt.tight_layout(); plt.show()
```

ETF	duration	dY 2022	predicted -D*dY	actual 2022
SHY	1.9	3.70%	-7.0%	-3.9%
IEF	7.5	2.37%	-17.8%	-15.2%
TLT	17.5	2.07%	-36.2%	-31.2%

Figure 4.3 · 2022 — the duration ladder in real life (indexed to 100)

All three fall through the year, and the drawdown deepens with duration



Source: Author calculations, from the article's computed results.

The ranking is exactly as duration predicts — and the gaps between predicted and actual returns are the tutorial's best teaching moment: coupon carry, convexity (helping TLT), and the fact that ETFs hold rolling portfolios rather than a single bond all show up in the residual.

4.4 Sanity check: how good is the Taylor approximation?

Duration-only vs duration+convexity vs exact repricing, for the 30y bond across shocks up to ± 300 bp.

```
T = 30
p0 = bond_price(100, 0.04, 0.04, T)
d, c = duration_convexity(100, 0.04, 0.04, T)

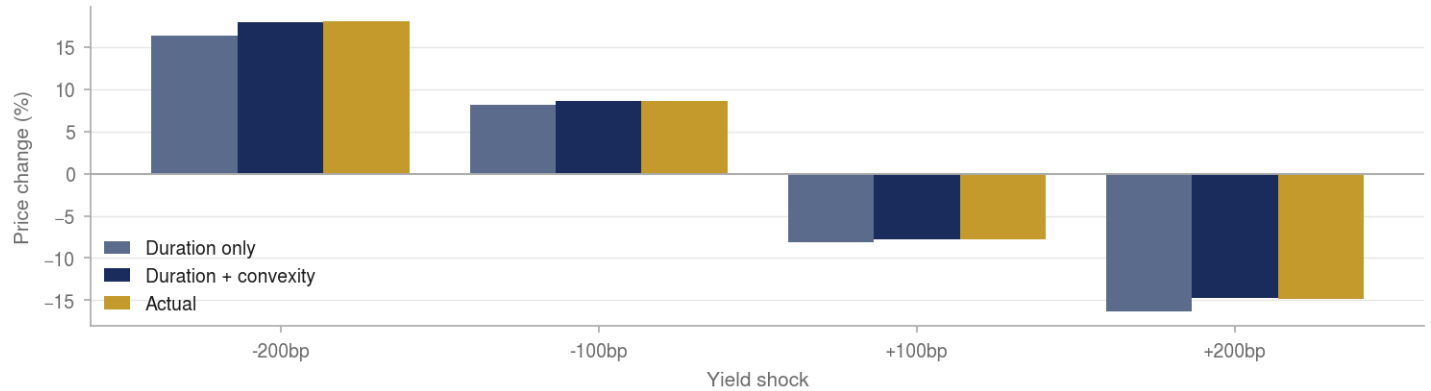
shocks = np.linspace(-0.03, 0.03, 61)
exact = np.array([bond_price(100, 0.04 + s, 0.04 + s, T) / p0 - 1 for s in shocks])
lin = -d * shocks
quad = -d * shocks + 0.5 * c * shocks**2

fig, ax = plt.subplots(figsize=(10, 4.5))
ax.plot(shocks * 100, exact * 100, "k", label="Exact repricing")
ax.plot(shocks * 100, lin * 100, "--", color="steelblue", label="Duration only")
ax.plot(shocks * 100, quad * 100, "--", color="darkorange", label="Duration + convexity")
ax.set_xlabel("Yield shock (%)"); ax.set_ylabel("Price change (%)")
ax.set_title(f"30y bond: Taylor approximation quality (D={d:.1f}, C={c:.0f})")
ax.legend()
plt.tight_layout(); plt.show()

s = 0.02
print(f"+200bp: exact {bond_price(100,0.04,0.06,T)/p0-1:+.2%}   dur-only {-d*s:+.2%}   dur+conv {-d*s+0.5*c*s*s:+.2%}")
```

Figure 4.4 · 30y bond — Taylor approximation quality

Duration alone overstates the loss and understates the gain; adding convexity closes most of the gap to the exact repricing



Source: Author calculations, from the article's computed results.

5. Conclusion

5a. Strengths

- **One framework, any bond** — price, duration and convexity summarise rate risk in three numbers
- **First-order accuracy is excellent** for small yield moves — the industry's daily risk language
- **Convexity correction** keeps the approximation honest even for $\pm 200\text{bp}$ shocks
- **Model-free verification** — ETF drawdowns in 2022 line up with durations, straight from public data

5b. Weaknesses & Limitations

- **Parallel-shift assumption** — duration assumes the whole curve moves together; real curves twist and steepen (key-rate durations fix this)
- **Constant yield reinvestment** — YTM assumes coupons reinvest at the same rate
- **Credit ignored** — this is a rates framework; spreads and default risk need their own layer

- **ETF $\neq$ single bond** — rolling portfolios have carry and roll-down effects the single-bond math misses

5c. Applications in Practice

- Portfolio duration targeting and immunisation
- Rate-shock stress tests (the $\pm 100\text{bp}$ tables in every risk report)
- Barbell vs bullet trades — convexity is the whole game

5d. Alternatives & Extensions

- **Key-rate durations** — sensitivity to individual curve points
- **Yield curve bootstrapping with QuantLib** — the natural companion piece (curriculum M2)
- **Term-structure models** — Vasicek, Hull-White, CIR for simulating rates rather than shocking them
- **The SVB collapse** — our future case study on what happens when duration risk meets deposit flight